

MID TERM EXAMINATION
B.TECH PROGRAMMES (UNDER AEGIS OF USICT)
Third Semester, November 2023

Paper Code: CIC-209**Subject: Data Structures****Time: 1½ Hrs.****Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1. (a)	What is a data structure? Discuss different operations performed on it.	2	CO1
(b)	Consider a two-dimensional array A[5:14][10:20] . Find the total number of elements in array A. Suppose Base Address (A) = 200 and there are w = 4 words per memory cell. Determine the address of A[10][10] using row-major order.	2	CO2
(c)	Explain the different Asymptotic notations (Big-O, Big-Ω, Big-Θ) used in algorithm analysis.	2	CO1
(d)	Explain sparse matrix and its triplet (3-tuple) representation with an example.	2	CO2
(e)	Construct a Max-Heap using the following data elements: 10, 20, 5, 30 . Show the heap structure after insertion of each element.	2	CO3
2. (a)	Convert the following infix expression into Reverse Polish Notation (postfix) using stacks: (A * B / (C ^ D)) + (E / F)	5	CO2
(b)	What is a B-Tree? What are the properties of a B-Tree? Construct a 3-way B-Tree (order 3) for the following elements: 10, 15, 20, 30, 5, 8, 40 .	5	CO2
3. (a)	What is a DEQUE? Explain different types of DEQUE. Write an algorithm for insertion at the front end in a DEQUE.	5	CO2
(b)	Create a Binary Search Tree (BST) for the following data inserted in sequence: 4, 6, 9, 13, 18 . What kind of BST is created? Is there any kind of problem with such kind of BST? If yes, give the solution for it.	5	CO2
4. (a)	Write an algorithm/function to delete an element from the end of a linear singly linked list.	5	CO2
(b)	The post-order and in-order traversals of a binary tree are given below: Post-order: D, F, E, B, G, L, J, K, H, C, A In-order: D, B, F, E, A, G, C, L, J, H, K Construct the corresponding binary tree showing all steps. Write its equivalent pre-order traversal and a recursive function/algorithm for pre-order traversal.	5	CO2

**** End of Question Paper ****